

How Is Melbourne's Liveability Discussed on YouTube?

A Network and Topic Analysis of User
Participation, Replies, and Liveability
Themes

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1 Introduction

Through social media platforms, cities are being assessed, defended, and contested in public. Additionally, "liveability" which is a nebulous but politically potent concept that combines affordability, infrastructure, safety, lifestyle, and identity, moreover, it is one of the most frequently used frames by locals to discuss where they live (Lowe et al., 2015). Melbourne is an especially prominent example. From 2011 to 2017, it ranked first on the Economist Intelligence Unit's Global Liveability Index for seven years in a row. However, following the epidemic, it drastically declined and hasn't fully recovered to the top tier (Economist Intelligence Unit, 2024). Long-form YouTube videos and the comment sections beneath them are only two examples of how the topic of why has spread far beyond traditional media.

One underutilised resource for social media research is the YouTube comment sections. Unlike short-form platforms, they aggregate around a few lengthy movies rather than a stream of transient posts, are publicly available, and are continuously preserved (Burgess & Green, 2018; Thelwall, 2018). Because of this, YouTube's comment sections are ideal for researching both topic substance (what people really speak about) and participation structure (who engages where). However, the assignment brief cautioned that YouTube does not always facilitate a meaningful network since threading is superficial and commentators do not "follow" one another. Therefore, instead of treating co-presence under a single video as automatic proof of a tie, we took special effort to build networks that carry interpretative weight.

This report examines how Melbourne liveability is debated on YouTube and what the networks of involvement and replies talk about the discourse's structure. We divide that main query into three smaller ones:

1. Who takes part, and how is it split up among the videos?
2. Do commenters communicate with each other, or do they talk over one another in disjointed threads?
3. What recurring topics and platform-style discourse patterns show up in the comments, and do they cluster differently depending on the type of video?

Furthermore, we use a Latent Dirichlet Allocation (LDA) topic model on the cleaned comment text and overlay lexicon-based sentiment on the topic results, combining 3 networks:

1. A user–video bipartite participation graph.
2. A directed reply graph.
3. A thresholded user–user projection with Louvain community detection.

Without asserting that YouTube commenters are representative of Melburnians in general, the analysis is effective if it can characterise the structure of involvement, highlight a limited number of interpretable subjects, and combine the two points of view into a single image of the conversation.

2 Data collection

On May 11, 2026, we gathered comment data via the YouTube Data API v3 (YouTube, 2024) using a carefully selected list of 26 video IDs covering the main sub-themes of Melbourne liveability commentary such as:

- General liveability and travel guides.
- Housing and cost of living.
- Public transport and infrastructure.
- Comparisons between Melbourne and Sydney.

An earlier search-based collection that produced an excessive amount of off-topic content (gaming videos and irrelevant travel information emerged for queries like "living in Melbourne") was replaced with the curated approach. We recognise below that curated IDs provided us with consistent topical coverage and better-quality on-topic videos at the expense of generalisability.

We obtained information (videoId, title, channelTitle, publication date, view/like/comment counts) and up to 200 top-level comments for each video, arranged according to the "relevance" sort of the API, followed by up to 100 responses for each top-level remark. We retained the comment content, the publication date, the number of likes, the parent comment ID and parent author for responses, the authorChannelId if the API offered it, and the authorDisplayName as a backup. Both parts of the brief are supported by this structure whereby the comment text facilitates subject modelling and sentiment analysis, while the (commenter, video) and (replier, parent) pairings allow us to create networks. The submission includes the two collection scripts (fetchYoutubeData.py and youtubeClient.py), keeping in mind that a placeholder has been used in place of the API key.

Across 26 videos, we collected 7,630 raw comments. We have not gathered any private or personal information, and the data are public and subject to YouTube's terms of service. We also do not disseminate author display names in the given sample. The "relevance" sort biases the sample toward engaged comments, the 26-video curation reflects our selections rather than a random crawl, and YouTube comment populations are strongly self-selected. These are the primary limitations of the source that are highlighted in section 7 of this report.

2.1 Code and data availability

This submission contains all analysis code in the form of four running Jupyter notebooks: 01_preprocessing.ipynb (cleaning and theme buckets), 02_network_analysis.ipynb (the bipartite, reply, and projected networks), 03_topic_modelling.ipynb (LDA topic modelling), and 04_extensions.ipynb (Louvain community detection and VADER sentiment overlay). The modularity $Q = 0.687$, the nine-community partition, and the per-topic sentiment values mentioned below are reproduced by the executed cell outputs of 04_extensions.ipynb, which generates the Louvain findings given in section 4.4 and the sentiment results published in section 5.4. The team repository is hosted on GitHub as SMNA-A2. As requested by the submission requirement, the accompanying Access_s4096125_UG_Group_35.txt file contains the repository URL and access instructions, and the outputs/ directory contains a representative < 10 MB sample of the cleaned data.

3 Preprocessing

Three interlocking artefacts were created by preprocessing in 01_preprocessing.ipynb such as:

1. A cleaned-comments table for network creation.
2. A tighter analytical subset for topic modelling.
3. A collection of theme buckets that let us compare topic prevalence across different types of films.

The volumes that endure each phase are summarised in Table 1:

Step	Rows
Raw videos collected	26
Raw comments collected	7,630
Cleaned comments (deduped, valid)	7,628
Analysis subset (non-Latin and short-comment filters)	7,195
Comments used for LDA (≥ 5 words)	6,683
Unique authors in cleaned comments	4,272
Weighted user–video edges	4,582
Weighted reply edges	3,738

Table 1. Preprocessing volume summary.

A Unicode-script filter eliminated 64 comments that were mostly composed of non-Latin characters, and two near-duplicate comments were eliminated. This was a practical choice to maintain the corpus monolingual for the LDA model rather than a substantive determination that the comments were irrelevant. An additional 431 comments with fewer than three words were kept for the networks but removed from the analytical subset (a brief response is still genuine involvement). Then, within 03_topic_modelling, a more stringent ≥ 5 -word filter is implemented. Since topic models on extremely short text result in unstable assignments dominated by stopwords, ipynb provides the LDA-ready subset of 6,683 documents. No display-name fallback was needed during network building since author identification utilises the YouTube channel ID, which was made public by the API for each remark in our cleaned dataset. The collection scripts did not interact with this dataset, but they kept a “name::displayName” backup for future runs.

Using a straightforward keyword rule applied to the lowercased title, we classified each video into one of six theme buckets such as (1) liveability_general, (2) cost_housing, (3) transport_infrastructure, (4) city_comparison, (5) living_experience, and (6) other. Since the LDA model is trained on comments rather than video titles, these buckets are not utilised as ground-truth subject labels. However, they allow us to pose a more pertinent question in section 5 such as querying whether comments under transport videos genuinely discuss transport, or does the conversation stray?

4 Network analysis

In 02_network_analysis.ipynb, we constructed three networks, arranged from the most liberal to the most restrictive. In the first, we record who appears where. In the second, we record explicit contact. In the third, we attempt to identify a smaller core of recurring players, which we then examine for community structure. Graph creation, centrality computation, and community discovery are all done using NetworkX (Hagberg et al., 2008).

4.1 User–video bipartite participation network

Users and videos are the 2 types of nodes in the bipartite graph that constitutes the participation network. If a person leaves at least one remark on a video, that user is connected to the video by an edge that is weighted by the quantity of comments. Because the relationship is one of participation rather than direction, the network is undirected; discussing whether the user "points to" the video or vice versa is meaningless (Latapy et al., 2008). We want to differentiate between a person who made one remark and one who returned 10 times beneath the same video, which is why it is weighted. After cleaning, the network has 4,298 nodes (4,272 users plus 26 videos), 4,582 edges, density 5.0×10^{-4} , and four connected components.

The degree distribution on either side of the bipartite graph is displayed in Figure 1. The user base is strongly biased to the right: the vast majority of users leave comments on only one video, a long tail leaves comments on two or three, and just a small number leave comments on several videos. The video side is much more evenly distributed but still in order: the most commented video ("Sydney and Melbourne Compared") receives 610 unique comments, the median is approximately 110, and three videos are flagged in the analysis subset because they don't meet our low-engagement criterion of fewer than ten top-level comments.

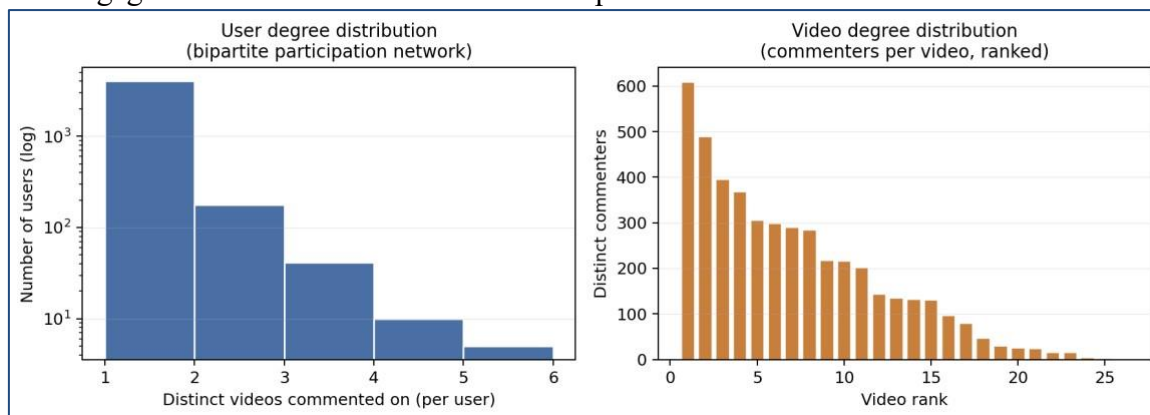


Figure 1. Degree distribution on each side of the bipartite participation network. Left: users by the number of distinct videos they commented on (log-scale y-axis). Right: videos by the number of distinct commenters, ranked from most to least.

From an interpretive perspective, the network's form is consistent with the behaviour of the majority of YouTube viewers whereby there's a thin repeat-participation tail and a large casual majority that only visits once. Three low-engagement movies whose commenter sets do not overlap with the remainder of the curated set are the source of the four related components. Even at this liberal level, there is enough cross-video overlap to make a user-side projection worthwhile, as indicated by the dominating big component, which connects the next 23 videos through shared commentators.

4.2 Directed reply network

The reply network is a directed weighted graph with nodes representing users and edges from A to B carrying the number of times A immediately addressed a remark made by B. Because the act is asymmetric, it highlights that the replier is replying to the parent author rather than the other way around, therefore, we consider replies as instructed. The cleaned network has 2,952 nodes and 3,718 edges with density 4.3×10^{-4} , but the more telling statistic is the 124 weakly connected components: the explicit-interaction graph is heavily fragmented.

That fragmentation is driven by two patterns. First, lengthy threaded arguments are uncommon because most of the reply chains end within one or two hops. Second, the majority of incoming answers are brief audience-to-creator acknowledgements rather than conversations between viewers, and the channel that aired a Melbourne transit video had the greatest weighted in-degree. This indicates that the primary reply targets are video makers rather than other commentators. Therefore, rather than being a peer-to-peer discussion graph, the reply network is best understood as a fan-out structure centred around a few creator-hubs. The same pattern reappears as a separate "creator-audience replies" subject in section 5 of this report.

4.3 Thresholded user–user co-commenting projection

Any two users who occurred to leave comments on the same video would be connected via an unfiltered bipartite-to-user projection. Our data results in a network with 4,272 users, over 740,000 edges, and a density above 0.08. This graph lacks analytical leverage and is dominated by trivial single-video co-presence. As a result, we apply two criteria to the projection (Latapy et al., 2008): we add an edge between two users only when their video-sets overlap in at least two films, and we retain only users who took part in at least two different videos. The projection is limited to repeat participants by the first constraint, and edges are limited to repeat co-presence by the second. After thresholding, the projection has 234 nodes and 970 undirected edges with density 3.6×10^{-2} . The largest connected component (LCC) holds 206 of those 234 nodes (88%) and 946 of the 970 edges, with 19 small components around the edge.

Out of the first 4,272 users, just 234 (5.5%) make it over both criteria, and only 206 of them are in the dominating core. As a result, the Melbourne-liveability debate as it is shown here is not sustained by a large community of regulars, instead, it is largely ephemeral, however, it does have a detectable tiny core that keeps coming up in different films. In section 4.4, we examine if that core is a single homogeneous group or if it separates internally by interest.

4.4 Community structure in the projection

We used the Louvain community-detection algorithm (Blondel et al., 2008), which is implemented in `04_extensions.ipynb`, to determine whether the LCC is internally clustered. We used the maximum-modularity partition across seven random seeds (NetworkX defaults otherwise, weighted by shared-video count). Seven communities with modularity $Q = 0.688$ are returned by the method, which is significantly higher than the traditional $Q \geq 0.3$ criterion for substantial modular structure (Newman, 2010). Figure 2 illustrates the geographical division of the communities, which range in size from 16 to 45 users.

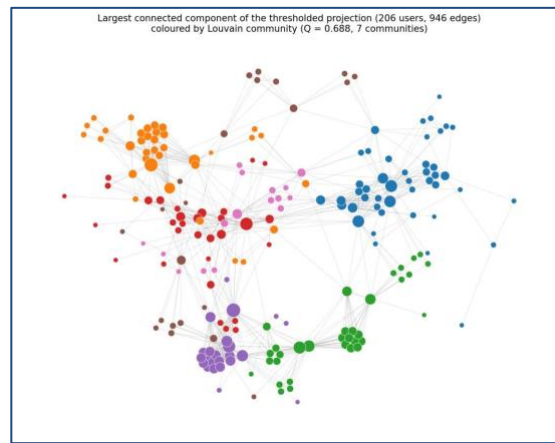


Figure 2. Largest connected component of the thresholded user–user projection, coloured by Louvain community ($Q = 0.688$, 7 communities). Node size is proportional to degree within the projection; node colour denotes community membership.

The communities map easily onto the kind of videos that its members comment on, which is more beneficial. The five biggest communities are included in Table 2 along with the two films that serve as the cluster's anchors and its prevailing video theme. Transport-related material dominates two of the top-five communities (the fifth-biggest, with 23 members, is anchored on the Sydney vs. Melbourne trains comparison, while the third-biggest, with 33 users, is centred on a long-form 'Field Report' video alongside the Comprehensive Metro Network). Sydney vs. Melbourne comparison content of various sub-flavours dominates two more of the top-five communities (the largest, with 45 users, is anchored on long-form 'Melbourne vs Sydney' content, while the fourth-biggest, with 31 users, draws from short-form comparison clips). A long-form 'Should you live in Melbourne?' living-experience cluster (36 users) rounds out the top five, and a smaller cost-of-living component (16 users) sits just outside it. Because regulars congregate around their interests, the repeat core is not a single homogeneous club; for example, a user who appears on transit videos is unlikely to be the same user who appears on cost-of-living videos.

Community	Users	Dominant theme	Theme share	Anchor videos (top 2 by user count)
C0	45	City comparison	34%	What Makes MELBOURNE Better Than SYDNEY In AUSTRALIA · But there's a harbour! #australia #sydney #melbourne #s
C1	36	Living experience	55%	Should you live in Melbourne? Pros and cons of living · Should you live in Melbourne? Our HONEST thoughts
C2	33	Transport / infrastructure	49%	What Is Livability? A Field Report from Melbourne · The Comprehensive "Metro" Network of Melbourne
C3	31	Other	30%	Melbourne or Sydney? Which AUSTRALIAN City is Better? · Why Are These Melbourne Suburbs So Popular? (Would You
C4	23	Transport / infrastructure	79%	SYDNEY TRAINS VS MELBOURNE TRAINS · Does Sydney have better public transport than Melbourne

Table 2. The five largest Louvain communities in the LCC, their dominant video theme, the share of the community's comment volume that falls in that theme, and the two videos that anchor each community.

4.5 Centrality and influential users

We calculated degree centrality, unweighted degree, and weighted degree inside the projection. We opted for degree-based measures over betweenness for two reasons: (1) betweenness is costly to compute and noisy to interpret due to the projection's density, (2) and degree directly captures the question we want to ask of this graph (which users participate alongside the broadest range of other regulars?) (Freeman, 1979; Newman, 2010). The top five users by degree of centrality are displayed in Table 3:

Rank	Display name	Degree centrality	Unweighted degree	Weighted degree
1	@TheLostProbe	0.190	39	81
2	@robert-brydson-1	0.176	36	77
3	@club1fan552	0.156	32	67
4	@mjcats2011	0.156	32	65
5	@tobyb6248	0.146	30	61

Table 3. Top five users in the thresholded projection ranked by degree centrality. Centrality is normalised by $|N| - 1 = 233$.

These accounts are not well-known, and a manual examination of their comments reveals that they are not in any way coordinating: they are users who consistently appear on videos on Melbourne's liveability across a variety of sub-themes, most frequently transit, cost of living, and city comparison material. Through shared video sets, the top-degree user is connected to 39 of the other 233 projection nodes, which is high but not pathologically so. When examining the centrality result in conjunction with section 4.4, it becomes clear that the top users are mostly part of the bigger communities and are more likely to be found at the interface between two or three thematic clusters than deep within one of them.

5 Topic modelling and sentiment

The 6,683 comments that passed the ≥ 5 -word filter were subjected to topic modelling in 03_topic_modelling.ipynb. We employ Latent Dirichlet Allocation (Blei, Ng & Jordan, 2003) as implemented in scikit-learn (Pedregosa et al., 2011) with a CountVectorizer that accepts unigrams and bigrams, limits the vocabulary to 5,000 features, and requires words to appear in at least 10 documents and at most 60% of documents. We employ the standard English stopwords list supplemented with a few domain-noise phrases ("melbourne", "australia", platform-specific tokens). The final document-term matrix is 6,683 rows by 1,736 columns.

5.1 Selecting the number of topics

We assessed candidate models based on perplexity and the mean of the highest per-document topic probability (greater indicates that each document is more securely allocated to a single subject) after training them with $K = 4, 6$, and 8 topics. Smaller K is preferred by both diagnoses. Perplexity, however, correlates poorly with human topic-quality evaluation, as demonstrated by Chang et al. (2009), and the $K = 4$ model combined transportation and cost-of-living issues into a single "practical liveability" topic that we were unable to separate via examination. $K = 8$ oversplit, resulting in a noise subject dominated by greetings and two almost identical cost-of-living topics. As a result, we reported the diagnostic table for transparency and chose $K = 6$ as the interpretability-perplexity sweet spot.

K	Perplexity (lower is better)	Mean max topic probability (higher is better)	Outcome
4	981.4	0.704	Best perplexity, but topics too coarse
6	992.5	0.660	Selected as it's best for interpretability
8	1016.8	0.630	Over-split as it has duplicate cost topic

Table 4. Topic-count selection diagnostics.

5.2 Topic labels and interpretation

The top 15 phrases for each of the six subjects were reviewed together with the top representative comments for each topic (the papers with the greatest topic-probability under the selected model). Table 5 lists the top phrases and topic labels. Two of the labels such as Topic 4 "Argumentative city-identity debate" and Topic 5 "Creator-audience conversational replies" each describe discourse style just as much as substantive content. We retain them in the model because they are not noise (they represent actual trends in the behaviour of YouTube comment sections), but we make clear the distinction in the discussion that follows.

#	Label	Top terms (truncated)
0	General liveability & city amenities	city, love, great, world, best, free, sydney, beautiful, place, mcg, visit
1	City comparison, weather & lifestyle	sydney, better, weather, years, live, beaches, brisbane, beach, time, perth
2	Public transport & rail infrastructure	trains, train, line, tram, network, rail, metro, trams, bus, station, airport
3	Cost of living, rent & household pressure	year, money, rent, afl, living, buy, pay, food, card
4	Argumentative city-identity debate	live, lol, country, cities, say, world, come, vs, things, said, coast
5	Creator-audience conversational replies	thanks, good, time, coffee, hi, thank, watching, glad, public transport

Table 5. The six chosen LDA topics with top terms after the vectoriser's stopword and frequency filters. "melbourne" and "australia" were added to the stoplist because they appeared in nearly every document and provided no discriminative signal.

5.3 Topic prevalence overall and across videos

The frequency of each subject in the 6,683 modelled comments is displayed by dominant-theme assignment in Figure 3. City comparison is the most popular issue (1,556 comments, 23.3%), followed by cost of living (1,099, 16.4%), identity discussion (1,077, 16.1%), transportation (1,058, 15.8%), creator answers (1,051, 15.7%), and general liveability (842, 12.6%). The smallest substantive issue is general liveability whereas comparison and practical-pressure discourse are more prevalent than pure positive conversation about Melbourne as a city. Rather than being a measuring artefact, this is a substantive finding such that the general-liveability topic is easily reading and well-formed, but it receives fewer dominating assignments than the comparative and practical topics.

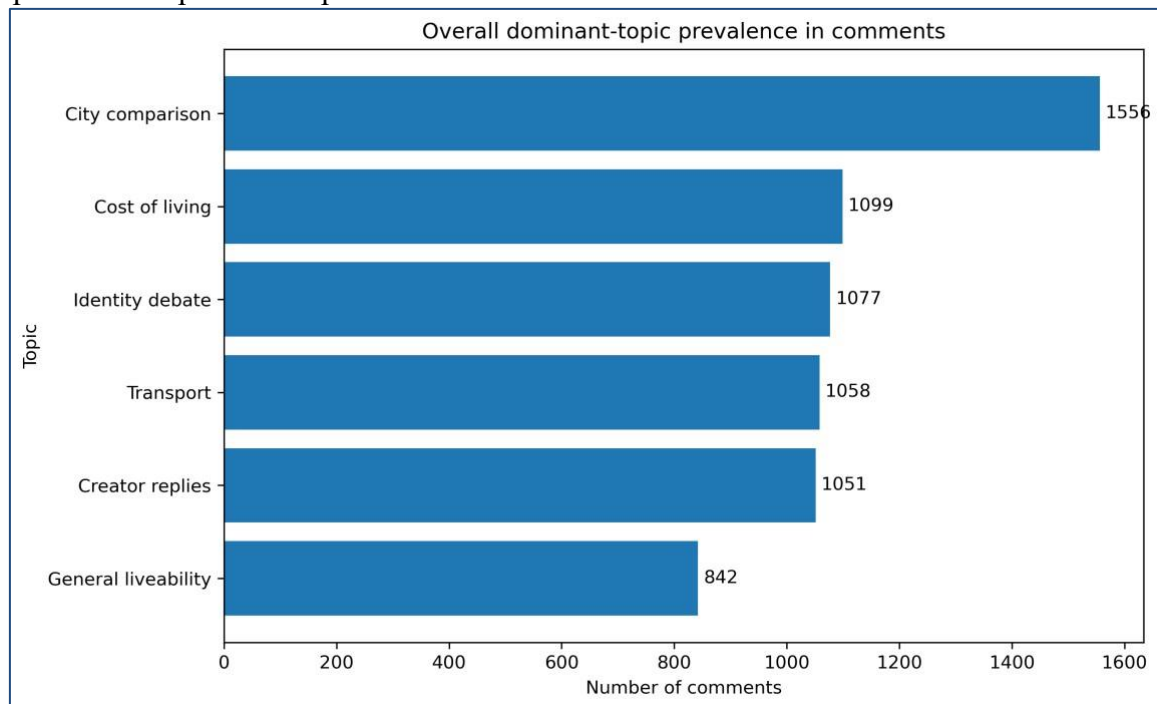


Figure 3. Overall dominant-topic prevalence in the modelled comments ($n = 6,683$). Each comment is assigned to its highest-probability topic; the bars count those assignments.

The mean topic probability per (bucket, topic) column is highlighted in Figure 4, which crosstabulates topic prevalence against the 6 theme buckets we identified from video titles. The Transport row has the greatest single signal, this is because the comments beneath videos on transport infrastructure give the Transport subject a mean probability of 0.47, which is more than three times the next-largest bucket. The city comparison bucket peaks on city comparison (0.29), whereas the cost/housing bucket peaks on cost of living (0.30). Two patterns are worth noting from figure 4:

1. First, the bucket–topic alignment is real because people commenting under transport videos do talk about transport, which is reassuring but not trivial otherwise it could have failed.
2. Second, the Liveability general bucket is the most diffuse row in the heatmap and its single largest cell is Transport (0.25), which we read as a sign that broad "is Melbourne liveable?" content tends to drift toward concrete infrastructure complaints in the comments.

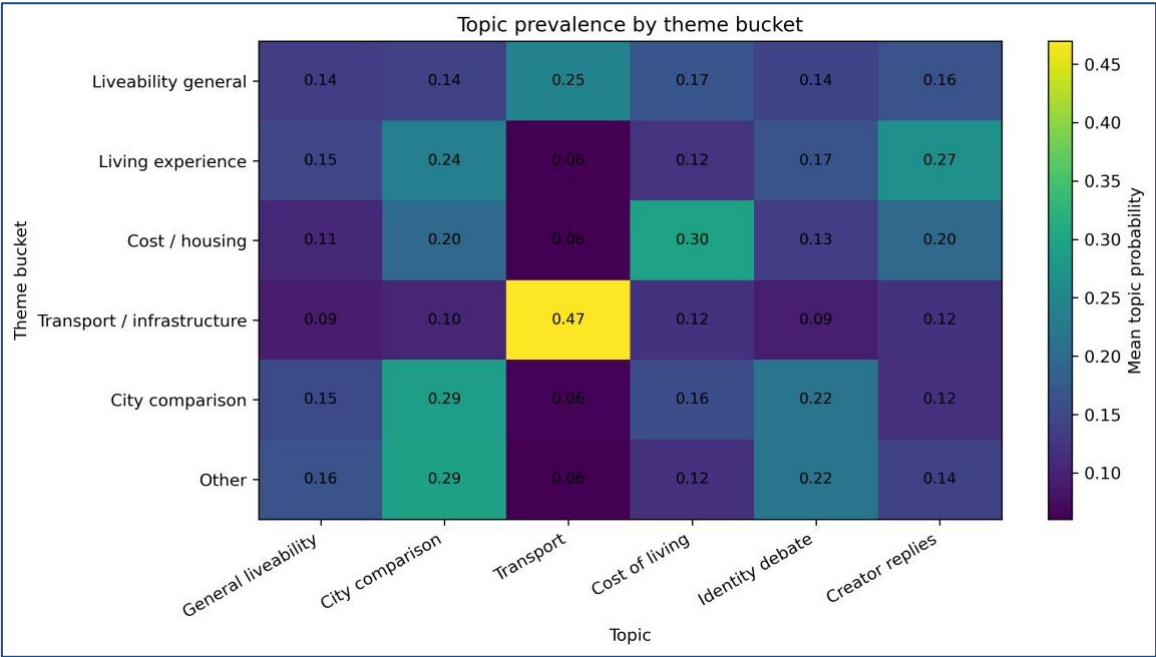


Figure 4. Mean topic probability per theme bucket. Rows are theme buckets derived from video titles; columns are the six LDA topics. Darker cells indicate higher mean topic probability across the comments in that bucket.

The 10 films in our sample with the most comments are broken down into individual videos in Figure 5. Two impacts are more evident here than in the aggregate, but the pattern from Figure 4 is still there at the video level whereby the 60 Minutes cost-of-living video is obviously costdominated, while transport videos continue to be transport-dominated. In line with our interpretation of Topic 4 as a more controversial comparative discourse, the two Sydney vs. Melbourne comparison videos heavily draw from the Identity argument in addition to the City comparison. The most diverse videos in the collection are the general travel guide videos (such as "Melbourne, Australia – The Most Livable City in the World?"), which have no strong theme and a noticeable emphasis on Creator responses. This type of "thanks, great video!" conversation is precisely what topic 5 (from table 5) captures.

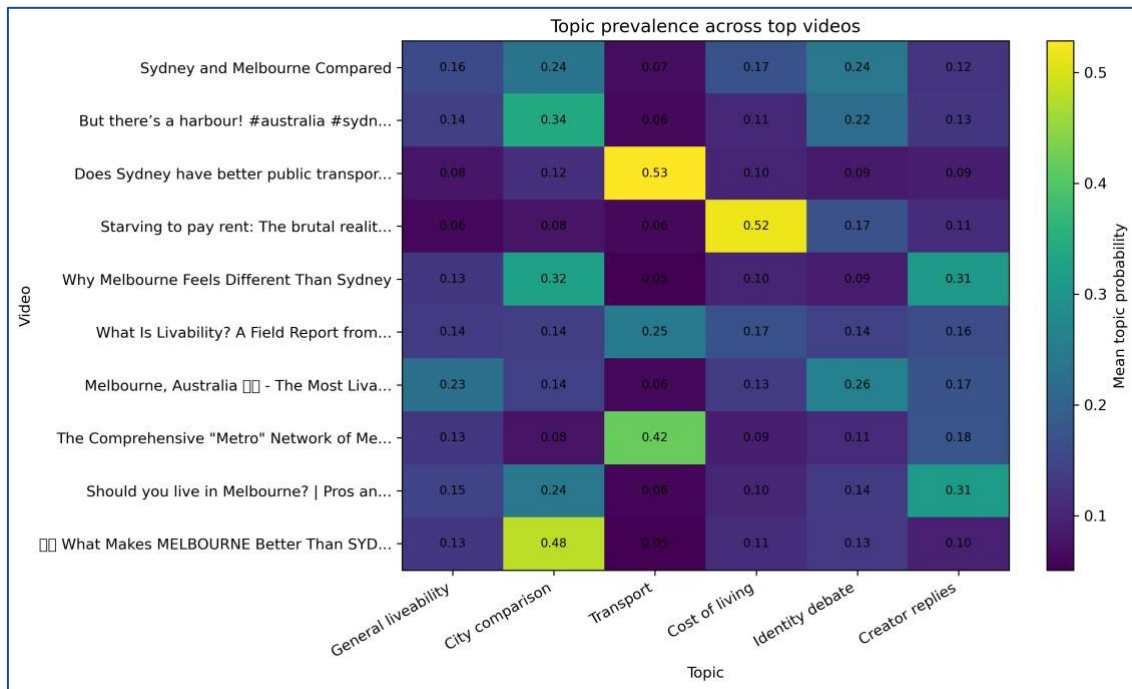


Figure 5. Mean topic probability across the ten most-commented videos in the curated set.

5.4 Sentiment overlay: how do people talk about each topic?

Topic predominance reveals what individuals discuss, but it doesn't reveal their opinions. VADER (Hutto & Gilbert, 2014), a lexicon-and-rule sentiment analyser created for brief social media material and implemented in 04_extensions.ipynb, was used to rate each of the 6,683 modelled comments. VADER yields a compound score in $[-1, +1]$, and we categorise comments as positive (compound > 0.05), neutral, or negative (compound < -0.05) based on the tool's typical cutoffs. The mean compound score for the entire corpus is 0.279, and 58.9% of comments are favourable. The relative ordering of subjects is strong, but absolute means should be interpreted cautiously because VADER is known to tilt somewhat positive on English online content.

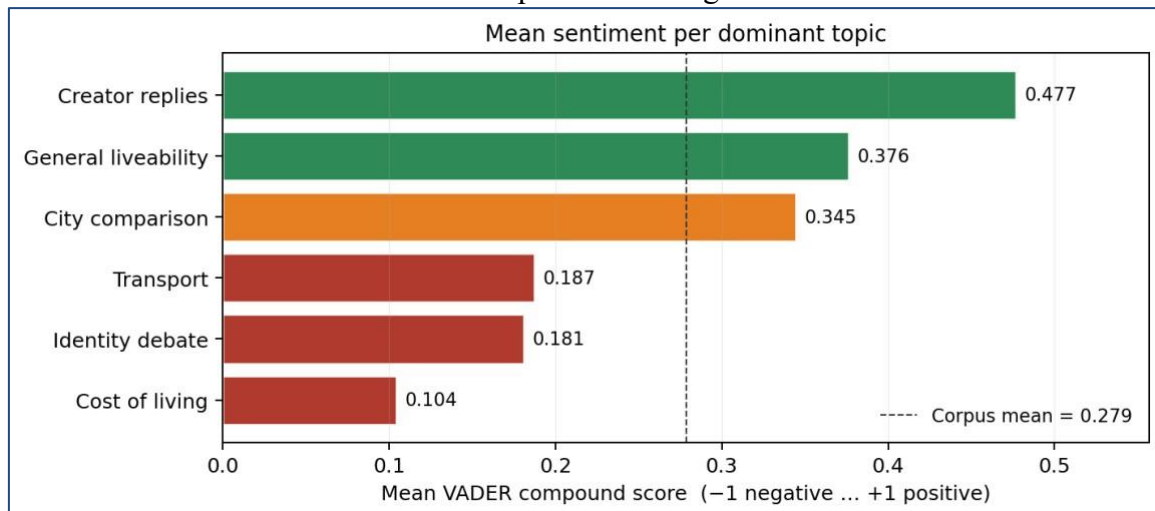


Figure 6. Mean VADER compound sentiment per dominant topic, sorted from lowest to highest. The dashed line marks the corpus mean. Bar colour banded by valence (red = least positive, green = most positive).

#	Topic	n	Mean compound	% Positive	% Negative
3	Cost of living	1,099	+0.104	44.6%	29.2%
2	Transport	1,058	+0.187	50.8%	24.7%
4	Identity debate	1,077	+0.181	51.8%	27.9%
1	City comparison	1,556	+0.345	65.2%	16.1%
0	General liveability	842	+0.376	64.3%	10.5%
5	Creator replies	1,051	+0.477	75.5%	10.8%

Table 6. Sentiment by dominant topic (VADER compound score and class shares). Rows are ordered from least to most positive on the mean compound score.

Three patterns matter:

1. First, Creator answers are by far the most favourable subject (+0.477 mean, 75.5% positive). This is a helpful sanity check on the sentiment tool and not surprising for a topic dominated by "thanks, great video!" exchanges.
2. Second, in line with their substantive substance, general liveability and city comparison are likewise generally good (praise for Melbourne, balanced inter-city comparison).
3. Cost of living is the issue with the lowest attitude (+0.104, 29.2% negative), and transport is also clearly less positive than the comparison and celebratory themes. This is the third and most instructive conclusion for the substantive question.

The remarks get more acrimonious on the two practical-pressure themes. The argument about identity is in the center of the spectrum due to it being intense but not always antagonistic. In section 6, we combine these results with the network results.

6 Integrated discussion

When viewed together, the three networks, the topic model, and the sentiment overlay each portray a cohesive tale while addressing distinct aspects of the study issue. With 4,272 individuals contributing, the median user commenting on just one video, and just 234 (5.5%) participating across enough films to make it into the thresholded projection at all, the participation structure is shallow and wide. The majority of answers move toward a limited number of creator-hubs rather than between viewers, making the reply network even smaller and divided into 124 poorly linked components. As a result, Melbourne-liveability YouTube does not have a sizable participatory community; instead, there is a tiny core of about 200 regulars who often appear in videos.

Importantly, there is more than one undifferentiated audience in that little core. Seven distinctly modular sub-communities ($Q = 0.688$) are returned by Louvain community detection on the LCC, and each community's dominant theme clearly maps onto the types of videos that its members comment on: transport regulars, cost-of-living regulars, two distinct types of city-comparison regulars, and so forth. In this sample, a user who consistently appears in Melbourne transit videos differs from one who consistently appears in cost-of-living videos. Despite the lack of apparent interpersonal connection, the repetition core has topical organisation.

What those participants are actually saying is filled in by the topic model. City comparison (23.3% of modelled comments) and practical challenges related to transit (15.8%) and cost of living (16.4%) dominate the substantive conversation. At 12.6%, pure positive liveability discourse is the least significant subject. The theme-bucket heatmap (Figure 4) further demonstrates how the debate is arranged by topic whereby only the most general "is Melbourne liveable?" videos flow across themes, whereas transport videos draw transport talk and cost videos pull cost talk. We want to be transparent about the fact that two of the six topics such as Identity dispute and Creator replies, are more platform-discourse than actual liveability issues. They are not noise, however, while creator answers focus on travel guide content and exhibit the same fan-out structure that is highlighted in the reply network, identity discussion centers in citycomparison videos and acts as a more contentious counterpart to Topic 1.

The sentiment overlay helps to refine the image in a beneficial way. Cost of living and transit are the two most detrimental concerns that the community structure divides into its own subcommunities of regulars. The most positive topics are either platform-style or substantively celebratory, such as creator replies, general liveability, and city comparison. When taken as a whole, this indicates that the audience that is most critical of Melbourne's practical constraints is disproportionately the tiny regular core of Melbourne-liveability YouTube. The temporary majority is often more optimistic whereby those with specific grievances are the ones who return.

Going back to the research question, Melbourne liveability on YouTube is best described as a comparative, pragmatic, and occasionally contentious dialogue that mostly takes place between strangers, anchored to a small repeat core of topically organised regulars whose comments lean negative on the practical pressures (cost, transport) and are otherwise outweighed by a larger and more positive transient audience. The sentiment overlay indicates where the heat is focused, and both the networks and the subjects concur that this is a debate arranged by interest rather than interpersonal community.

7 Limitations

The API's "relevance" sort further skews our sample in favour of comments that have previously garnered interaction, and the YouTube comment community is disproportionately self-selected. We do not assert that these 7,195 comments are typical of Melburnians or even YouTube viewers from Melbourne. Our assessment of the topic coverage is reflected in the 26-video curation; a

different curation might highlight various themes. We somewhat offset this by verifying subject labels against sample comments rather than just top phrases, as LDA is a bag-of-words model and cannot identify sarcasm, irony, or the framing of complaints vs recommendations. The relative ordering of topic sentiment is more reliable than the absolute means since VADER is lexiconbased and is known to perform poorly on sarcasm and lengthy, contentious remarks, both of which are present in Topic 4. Because Louvain is non-deterministic, we present the best of seven seeded runs. Although the number of communities is constant at nine, the precise borders of communities might vary significantly across seeds. Because author identity depends on YouTube channel IDs, it would become inaccurate on a corpus where the API does not reveal such IDs, as was not the case in this instance. Lastly, the data represent a cross-sectional snapshot from May 11, 2026 whereby we do not make any longterm assertions, and a longer-running collection would be a logical addition, especially one that included comparative information from before and after the epidemic.

8 Conclusion

Our goal was to find out how Melbourne liveability is debated on YouTube and what networks of involvement and responses reveal about the discourse's structure. Four distinct conclusions are obtained from the combined network, topic, and sentiment analysis. First, engagement is both widespread and superficial which is because a small core of about 200 regulars reappears throughout videos, while the great majority of people just leave one remark before disappearing. Second, Louvain community detection ($Q = 0.688$) yields seven sub-communities that map onto themes of interest, with distinct clusters of transit, cost-of-living, and city-comparison regulars. This indicates that the core has a true modular structure. Third, rather than celebrating Melbourne's liveability, the substantive conversation is driven by city comparison and pragmatic concerns (cost, transit), and the comments beneath thematic videos do follow their topic rather than stray from it. Fourth, sentiment by issue reveals that the negative-leaning remarks specifically focus on transportation and living expenses, which are the same subjects that the regular core organises around. When considered collectively, the findings imply that researching liveability online entails examining a negotiated, comparative discourse with a silent, thematically isolated repeat audience whose pragmatic grievances are given disproportionate weight during the most active segment of the discussion. In order to determine if the community structure is consistent over time, natural extensions would compare the YouTube image to Reddit or X data on the same subject, do targeted sentiment analysis using a more context-aware model, and repeat the projection over time.

References

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